

The sphygmomanometer consists of an inflatable bag attached to a cuff which can be fastened around the arm, a pump, a pressure gauge and a pressure release valve. Arterial blood pressure does not usually vary greatly with posture, but blood pressure readings are nevertheless normally made with the patient seated. The inflatable bag is placed on the inner surface of the upper arm, overlying the brachial artery, and is held in place by wrapping the cuff around the arm. If the pressure in the cuff is raised above the systolic pressure, the artery will be completely compressed and there will be no pulse and no blood flow downstream from the cuff. If the pressure is released, the pulse will return when the cuff pressure falls below the systolic pressure. If a stethoscope is placed over the artery, faint tapping sounds (the Korotkoff sounds) will be heard. As the pressure in the cuff falls to diastolic pressure, the sounds will become muffled. The standard procedure is to increase the pressure to about 30 mmHg above the point at which the pulse disappears, and then release the pressure at about 3 mmHg s⁻¹. The pressure at which the Korotkoff sounds are first heard is the systolic pressure, and the pressure at which the sounds become muffled is the diastolic pressure. After the diastolic pressure reading has been taken, the pressure is rapidly reduced to zero to restore the blood flow.

Several automated sphygmomanometers are available for non-invasive monitoring of blood pressure in intensive care units. The cuff is inflated at regular intervals by a motor driven pump, and the Korotkoff sounds are detected by a microphone. The signal from the microphone is suitably filtered to give a signal denoting the onset and muffling of the sounds. The cuff pressure, measured by a transducer, is recorded on an event recorder. In a fluid-filled system, the pressure transducer must be at the same level as the pressure which is being measured, so that the weight of the fluid column does not affect the reading. However, as the cuff is pressurized using air, which has a negligible weight, the height of the pressure transducer is not important. These instruments work adequately, but have difficulty in distinguishing between genuine Korotkoff sounds and sounds caused by patient movement.

Many other non-invasive methods of measuring blood pressure have been described in the literature, but none of them are now used clinically. Non-invasive methods are routinely used to measure the pressure within the eye (the intraocular pressure), and are described below. A similar technique can be used to measure intracranial pressure through the anterior fontanelle in infants.

18.3.1. Measurement of intraocular pressure

As an example of non-invasive pressure measurement, we will examine the measurement of intraocular (within the eye) pressure. Glaucoma (raised intraocular pressure) causes vision disturbances and ultimately blindness. Chronic glaucoma may not be noticed until the optic nerve is permanently damaged. The cause may be obstruction of the drainage of aqueous fluid from the anterior compartment of the eye. It is obviously important to know the normal range of intraocular pressure, and to be able to measure it non-traumatically. Unfortunately, the only definition of 'normal' for intraocular pressure is the pressure that does not lead to glaucomatous damage of the optic nerve head.

Numerous large studies of intraocular pressure have been made. The normal pressure is about 15 mmHg, with a standard deviation of about 3 mmHg. [Figure 18.1](#) shows the interpretation of the results as two populations (non-glaucoma and glaucoma) which overlap considerably. The standard deviation should also be treated with caution, as the distributions are skewed. The figures suggest a normal range of approximately 10–20 mmHg. The intraocular pressure is influenced by a number of factors (blood pressure, drugs, posture, blinking, etc) which are discussed in standard ophthalmology texts.

The problem, as far as we are concerned, is how we measure intraocular pressure non-invasively. The eye is a fluid-filled globe, and we can measure the pressure within by relating the deformation of the globe to an externally applied force. Tonometers (devices for measuring the pressure within the eye) are of two types: indentation and applanation (flattening). We will consider only the applanation type.

Applanation tonometers rely on the relationship between wall tension and pressure in an elastic sphere. The surface tension γ is related to the pressure difference ΔP across the curved wall by $\Delta P = 2\gamma/r$, where r

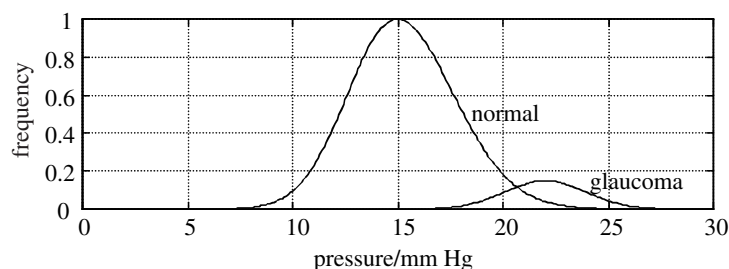


Figure 18.1. Distribution of intraocular pressure for normal and glaucoma groups.

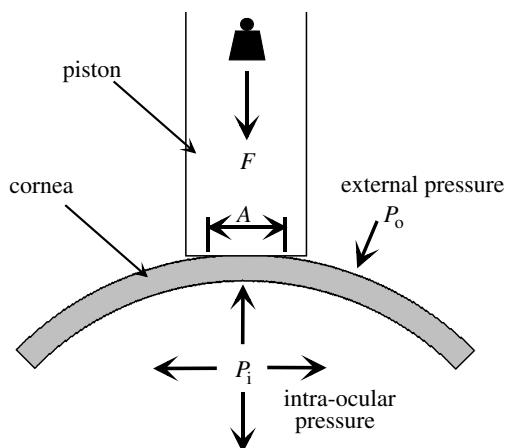


Figure 18.2. Applanation tonometer applied to the cornea.

is the radius. If the wall of the sphere is locally flattened (i.e. $r = \infty$), the pressure difference ΔP will be zero, and the pressure within the sphere can be related to the force applied to the wall by *pressure = force/area*. Ophthalmologists give this latter expression the grandiose name of the Maklakov–Fick or Imbert–Fick law, but it is no more than the standard definition of pressure.

The assumptions are that the eye is a spherical shell, perfectly flexible, has an infinitely thin wall, and that there are no surface tension effects due to fluids in contact with either surface of the wall. In practice, the eye violates all of these assumptions (see section 3.5.2 for the anatomy of the eye). The cornea has a central thickness of approximately 0.55 mm (figure 18.2), so that the outer contact area is less than the internal flattening. The wet surface gives a surface tension acting on the plunger, and a finite force is required to bend the cornea. Surface tension has a component acting in the same direction as the applied force, which is opposed by the bending force. In the literature, it is stated that it has been found empirically that these forces balance out when the contact area is 3.06 mm diameter. Given all the assumptions about the technique, and the wide range of intraocular pressures, this precision seems inappropriate. The resulting volume displacement is about 0.5 μl , compared to 180 μl for the volume of the anterior chamber and 60 μl for the posterior chamber, so that the applanation has a negligible effect on the intraocular pressure.

The Goldmann-type applanation tonometer applies a force to the surface of the cornea, and uses an optical system to determine when the applanated area is 3.06 mm in diameter. The cornea is anaesthetized

with a topical anaesthetic, and the tear film is stained with sodium fluorescein. The meniscus surrounding the applanated area is visible because of the fluorescence, and is split into two semi-circles by biprisms. The geometry is such that the inner edges of the semi-circles touch when the area applanated is correct. The intraocular pressure can then be measured directly from a scale on the tonometer. The thickness and curvature of the cornea can produce errors in the measurement.

Tonometers are now available that use a puff of air to deform the surface of the cornea. By measuring the velocity and volume of the air and the resulting deformation of the corneal surface it is possible to provide a reasonably accurate measurement of intraocular pressure in a very non-invasive manner.

18.4. INVASIVE MEASUREMENT: PRESSURE TRANSDUCERS

In general, greater accuracy will be achieved if the measuring device is inserted into the measurement volume, that is, if the measurement is invasive. The pressure transducer may be connected to the measurement volume by means of an in-dwelling air- or fluid-filled catheter, or may be mounted on the end of a catheter and inserted into the measurement volume. Intracranial pressure may be measured by placing a catheter in the ventricles, or by sub- or extradural transducers. The method of choice will be determined by cost, accuracy, frequency response, movement artefact and the time period over which monitoring will take place.

Placing the transducer itself within the measurement volume will give the best frequency response; freedom from movement artefact; no change in baseline pressure due to the patient moving relative to an external transducer; and will not require infusion of the catheter to maintain patency. However, catheter-tip transducers are expensive and fragile, and do not take kindly to repeated sterilization. If used for extended periods they may suffer from ingress of physiological fluids, and it is difficult to check the zero-pressure output of an in-dwelling transducer. Catheter-tip transducers which have only the diaphragm at the tip, with the diaphragm deflection measured externally through a fibre optic link, may overcome some of these problems.

The theoretically less desirable external transducer, with its fluid-filled catheter connected to the measurement volume, enables the technical problems of pressure measurement to be separated from the physiological problems. The problem of repeated sterilization destroying the delicate transducer has been overcome by physically separating the fluid from the transducer, so the transducer only requires, at most, surface decontamination. This has been achieved by including a flexible membrane in the transducer dome, which is in close apposition to the measuring diaphragm, and transmits the fluid pressure. The domes are sold as sterile disposable units. Indeed, this philosophy has been pursued to such an extent that sterile disposable transducers are commonly used.

Similar systems can be used for most fluid pressure measurements. The fluid-filled catheter system does suffer from movement artefacts and from apparent pressure changes if the position of the transducer changes relative to the measurement volume. Air-filled catheters overcome this problem, but are clearly limited to applications where air embolus is not a problem. One such area is the measurement of bladder function in ambulatory patients.

Intracranial pressure can be measured with an in-dwelling catheter, though this may provide a route for infection and cause trauma if any force is exerted on the catheter. Transducers are available which fit into a burr-hole, so that they are rigidly connected to the skull, thus reducing the risk of trauma. The actual pressure measurement may be made sub-durally, or, less invasively, by the transducer diaphragm being in contact with the external surface of the dura mater so that the dura mater remains intact.

Transducers

The most common type of pressure transducer consists of a diaphragm, one side of which is open to the atmosphere and the other connected to the pressure which is to be measured. Pressure causes a proportional

POLYGRAPH

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POLYGRAPH

- Psychophysiological measurements involve recording physiological responses that are associated with psychological processes such as emotions, stress, or anxiety.
- One of the most well-known instruments used for this purpose is the polygraph, commonly referred to as a “lie detector.”
- A polygraph records multiple physiological signals simultaneously to detect changes in the body that may occur when a person experiences stress or attempts deception.

POLYGRAPH

- Polygraph is widely used in forensic investigations, security screenings, and psychological research. It does not directly detect lies; instead, it measures physiological responses that may be associated with emotional or psychological stress.
- These responses are analyzed by trained examiners to determine whether a person's answers may indicate deception.
- Modern polygraph system was developed with contributions from researchers such as John Augustus Larson and Leonarde Keeler in the early 20th century. Their work helped establish the polygraph as a tool for studying physiological responses during questioning.

PRINCIPLE

- The polygraph works on the principle that psychological stress or emotional reactions produce measurable physiological changes in the body.
- When a person is anxious or attempting to conceal information, the autonomic nervous system may respond by altering heart rate, blood pressure, breathing pattern, and skin conductivity.
- The polygraph records several physiological parameters simultaneously.
- By comparing the responses to relevant questions and neutral questions, the examiner analyzes whether there are significant changes in the physiological signals.
- These changes may indicate emotional stress related to the question being asked.

PARAMETERS MEASURED

- A polygraph measures multiple physiological signals that reflect the activity of the autonomic nervous system.
- **Respiration Rate:** Breathing patterns are measured using pneumograph tubes placed around the chest and abdomen. Changes in breathing rate or depth may indicate emotional stress.
- **Blood Pressure and Heart Rate:** A blood pressure cuff or cardiovascular sensor records heart rate and blood pressure variations during questioning.
- **Galvanic Skin Response (GSR):** This measures changes in the electrical conductivity of the skin caused by sweating. Emotional stress can increase sweat gland activity, which changes skin conductance.
- **Body Movements:** Some modern polygraphs include motion sensors to detect body movements that might affect the readings.

COMPONENTS

- **Sensors and Transducers:** Various sensors measure respiration, cardiovascular activity, and skin conductance.
- **Amplifier and Signal Conditioning Unit:** The physiological signals are very small and require amplification and filtering before analysis.
- **Recording System:** Earlier polygraphs used paper chart recorders, while modern systems use computerized digital recording and data analysis.
- **Display and Analysis Software:** The recorded signals are displayed as waveforms on a monitor, allowing examiners to analyze the physiological responses during questioning.

WORKING PROCEDURE

- During a polygraph test, sensors are attached to the subject's body to measure physiological parameters.
- The subject is then asked a series of questions, which may include neutral questions, control questions, and relevant questions related to the investigation.
- As the subject answers each question, the polygraph records changes in respiration, heart rate, blood pressure, and skin conductance.
- The examiner compares the physiological responses to different types of questions.
- If significant physiological changes occur during certain questions, it may indicate emotional stress or possible deception.
- The results are interpreted carefully by trained examiners, considering all physiological data.

APPLICATIONS

- Polygraphs are widely used in criminal investigations to assist in determining whether a suspect may be withholding information.
- They are also used in security screening for certain government agencies and in psychological research to study stress and emotional responses.
- In addition, polygraph techniques are used in studies related to human behavior, interrogation methods, and stress responses.

For the treatment of disorders involving the various senses, especially those related to communication, a number of specialized fields have evolved. The *audiologist* determines deficiencies in the acuity of hearing, which often can be improved by the prescription of hearing aids. The *speech pathologist* treats disorders of speech, which may be due to damage to the structures involved in the formation of sounds or may have a neurological cause. The *ophthalmologist* is a physician who specializes in disorders of the eye, whereas the *optometrist* has no medical training and treats only those visual disorders that can be corrected by the prescription of eyeglasses. For the measurement of the acuity of the senses, as well as for the study of behavior, large numbers of instruments have been developed, which can be highly specialized.

The results of behavioral studies seldom show a simple cause-and-effect relationship but are usually in the form of statistical evidence. This peculiarity requires large numbers of experiments in order to obtain results that are statistically significant. As a result, especially in animal experiments, automated systems are frequently used to control the experiment automatically and record the results. The diversity of the field, on the other hand, has resulted in commercially available instruments that are often in the form of modules and building blocks which can be assembled by the experimenter into specialized systems to suit the requirements of a particular experiment.

One obvious way to study behavior is to measure the electrical signals in the brain and the nervous system that control the behavior, as discussed in Chapter 10. However, because the voltages recorded on an electroencephalograph are the result of many processes that occur simultaneously in the brain, only events that involve larger areas of the brain, such as epileptic seizures, can be readily identified on the EEG recording. For this reason, mental disorders generally cannot be diagnosed from the electroencephalogram, although the EEG is usually used to rule out certain organic disorders of the brain (e.g., tumors), which can show symptoms similar to those of nonorganic types of mental illness. The instrumentation used to measure the EEG is described in Chapter 10.

11.1. PSYCHOPHYSIOLOGICAL MEASUREMENTS

As stated in Chapter 10, many body functions, including blood pressure, heart rate, perspiration, and salivation, are controlled by the autonomic nervous system. This part of the nervous system normally cannot be controlled voluntarily but is influenced by external stimuli and emotional states of the individual. By observing and recording these body functions, insight into emotional changes that cannot be measured directly can be obtained. A practical application of this principle is the *polygraph* (colloquially

called the “lie detector”), a device for simultaneously recording several body functions that are likely to show changes when questions asked by the interrogator cause anxiety in the tested person.

For the measurement of blood pressure, heart rate, and respiration rate in psychophysiological studies, the same instruments are used as are utilized for medical applications (see Sections 6.1 and 6.2, and Chapter 8, respectively). For measuring variations in perspiration, a special technique has been developed. In response to an external stimulus, such as touching a sharp point, the resistance of the skin shows a characteristic decrease, called the *galvanic skin response* (GSR). The baseline value of the skin resistance, in this context, is called the *basal skin resistance* (BSR). The GSR is believed to be caused by the activity of the sweat glands. It does not depend on the overt appearance of perspiration, however, and the actual mechanism of the response is not completely understood. The GSR is measured most readily at the palms of the hands, where the body has the highest concentration of sweat glands. An active electrode, positioned at the center of the palm, can be used together with a neutral electrode, either at the wrist or at the back of the hand. In some devices clips are simply attached to two fingers. Frequently, in order to increase the stability of the measurement, nonpolarizing electrodes, such as silver-silver chloride surface electrodes (see Chapter 4), are used with an electrode jelly that has about the same salinity as the perspiration. In order to minimize the polarization at the electrodes, the current density is kept below $10 \mu\text{A}/\text{cm}^2$.

Figure 11.1 shows a block diagram of a device that allows the simultaneous measurement, or recording, of both the BSR and the GSR. Here a current generator sends a constant dc current through the electrodes. The voltage drop across the basal skin resistance, typically on the order of several kilohms to several hundred kilohms, is measured with an amplifier and a meter that can be calibrated directly in BSR values. A second meter, coupled through an *RC* network with a time constant of about 3 to 5 seconds, measures the GSR as a change of the skin resistance of from several hundred ohms to several kilohms. The output of this amplifier can be recorded on a suitable graphic recorder. A measurement of the absolute magnitude of the GSR is not very meaningful. The change of the magnitude of the GSR, depending on the experimental conditions and its *latency* (the time delay between stimulus and response), can be used to study emotional changes. A polygraph for recording physiological functions, including GSR is shown in Figure 11.2.

Instead of the change of the skin resistance, the change of the *skin potential* has been used occasionally. This is actually a potential difference of between 50 and 70 mV that can be measured between nonpolarizing electrodes on the palm and the forearm and that also shows a response to emotional changes.

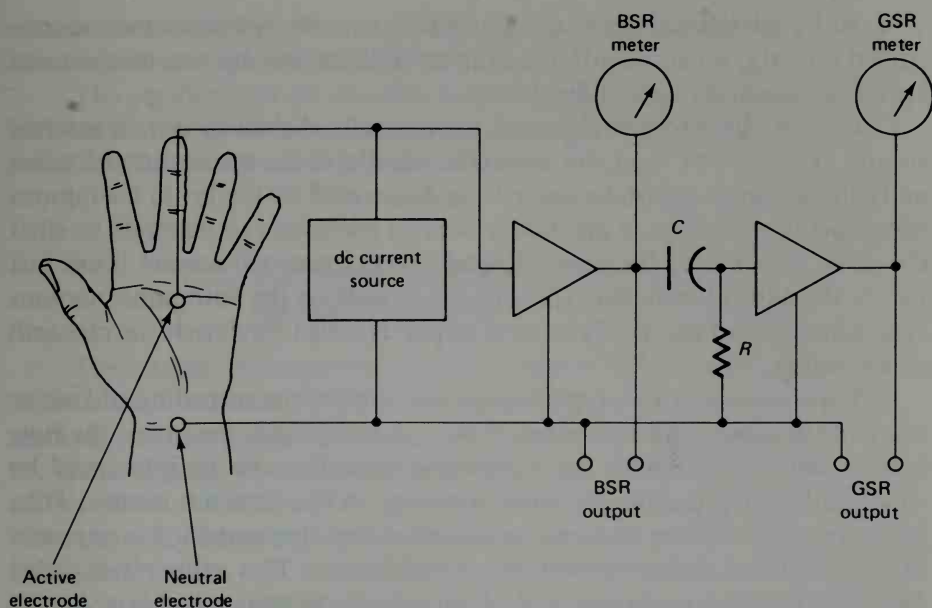
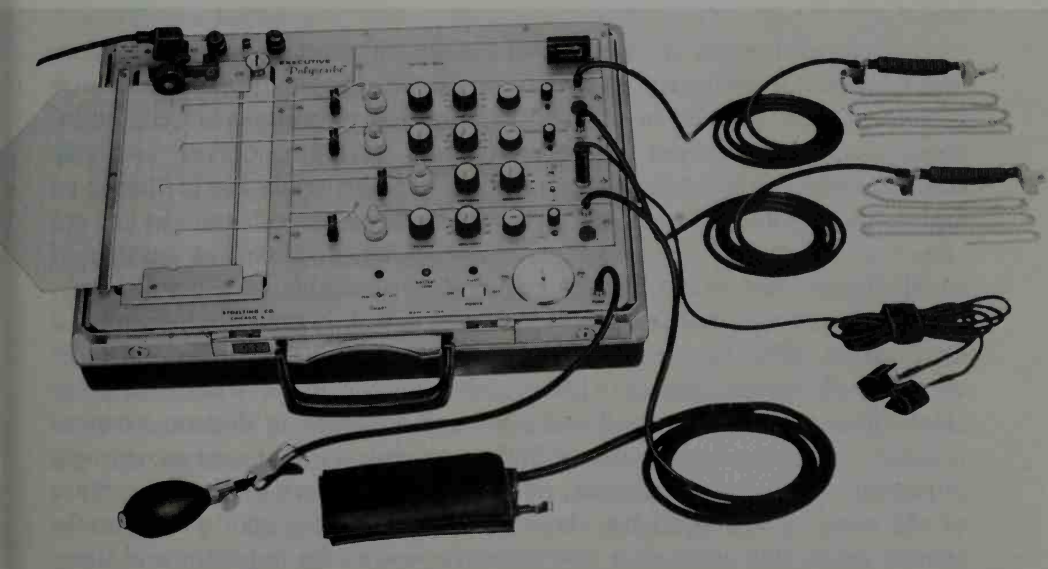


Figure 11.1. Block diagram of a device to measure and record basal skin resistance (BSR) and the galvanic skin response (GSR.)

Figure 11.2. Polygraph for the recording of four body functions. The sensors (right and bottom, clockwise) are for respiration (2 channels) galvanic skin response (GSR) and blood pressure changes. (Courtesy of Stoelting Co., Chicago, IL.)



Although the activity of the autonomic nervous system cannot be controlled directly, it can be influenced in an indirect way by two mechanisms known as *conditioning* and *feedback*.

Certain physiological responses are normally elicited by certain external stimuli. The view of food, for instance, stimulates the production of saliva and causes "one's mouth to water." As discovered by Pavlov in his famous experiments with dogs, a previously neutral stimulus can be made to elicit the same response as the view of food if it is presented several times just before the natural stimulus. This process of making the autonomic nervous system respond to previously neutral stimuli is called *Pavlovian* (or *classical*) *conditioning*.

Experiments of this type require the continuous recording of one or more of the autonomic responses. Pavlov, for example, measured the flow rate of saliva. Sometimes the autonomic responses can be influenced by simply informing the subject when a change in the response occurs. This, again, requires that the response be measured and that certain characteristics of it be signaled to the subject in a suitable way. This principle is called *biological feedback* or *biofeedback*. Although this technique had been known for some time, it received renewed interest during the early 1970s for possible therapeutic uses in controlling variables like heart rate, blood pressure, and the occurrence of certain patterns in the electroencephalogram. Biofeedback is described in more detail in Section 11.5.

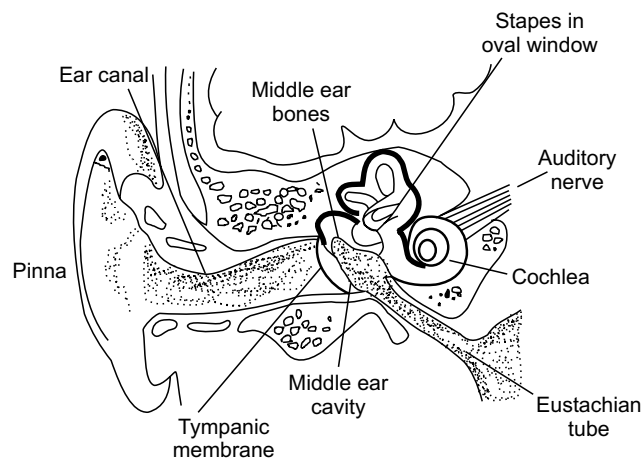
11.2. INSTRUMENTS FOR TESTING MOTOR RESPONSES

Motor responses, or responses of the skeletal muscles, are under voluntary control but often require a learning process for the proper interaction between several muscles in order to perform the response correctly. Numerous devices have been described in the literature, or are available commercially, to measure motor responses and to study the influence of factors like fatigue, stress or the effects of drugs. Some of these devices are very simple. Manual dexterity tests, for instance, consist of a number of small objects that the subject is required to assemble in a certain way, while the time required for completion of the task is measured. In related instruments called *steadiness testers* a metal stylus must be moved through channels of various shapes without touching the metal walls. An error closes the contact between wall and stylus and advances an electromechanical counter. The *pursuit rotor* uses a similar principle. A light spot moves with adjustable speed along a circular, or star-shaped, pattern on the top surface of the tester. The subject has the task of pursuing the spot with a hook-shaped probe that contains a photoelectric sensor. An indicator and timer

Audiometers and Hearing Aids

17.1 MECHANISM OF HEARING

Sound waves are longitudinal waves in which the motion of each particle of the medium in which the wave is travelling, moves backward and forward along a line in the direction in which the wave is propagated. The human aural system reacts to these oscillating pressure changes and transmits them to the brain through a series of steps. Fig. 17.1 shows the anatomy of the human ear. The outer ear consists of the pinna or auricle, together with the ear canal, the external auditory meatus, which is a convoluted tube, about 1 cm^3 in volume and terminates at its inner end in the tympanic membrane. The pinna scatters acoustic waves so that some of the scattered energy enters the auditory canal and pushes against the tympanic membrane during a wave of compression. The distance membrane moves is a function of the force and velocity with which the air molecules strike it and is, therefore, related to the loudness of sound.



► Fig. 17.1 *Anatomy of ear*

The tympanic membrane separates the ear canal from the middle ear cavity. The middle ear is exposed to atmospheric pressure only through the eustachian tube, which connects it to the pharynx and nose or mouth. The sound energy from the tympanic membrane is transmitted through the cavity of the middle ear, to the receptor cells in the inner ear, which are surrounded

by fluid. Thus, the major function of the middle ear is to transfer movements of the air in the outer ear to the fluid-filled chambers of the inner ear. A chain of three small, middle ear bones couple the tympanic membrane to a membrane covered opening, called the oval window. The total force on the oval window is the same as that on the tympanic membrane. The size of the window being very small, it experiences much greater force per unit area. One of the bones, called the stapes, rests upon the lower end of the cochlea and passes the vibrations directly into the fluid within.

The inner ear or cochlea is a fluid-filled coiled passage in the temporal bone. It is almost completely divided lengthwise by the basilar membrane. Most of the pressure wave received by the cochlea is transmitted to this membrane, which is deflected into the scale tympani. The membrane has different resonating properties along its length, responding to high frequencies at the stapes end and to low frequencies at its upper end. The membrane contains the sensitive receptor cells, which transform sound energy or pressure waves into action potentials. The nerve impulses thus initiated are propagated along the acoustic nerve fibres to the brain with a speed of 100 m/s. The pattern of nerve impulses arriving in the brain is associated with the subjectively experienced sound, which has attributes of loudness, pitch and timbre (quality).

The appreciation of sound is mainly a cerebral function. However, the recognition of notes is partly a function of the cochlea. Therefore, if it is defective, the individual may not hear certain tones.

Hearing is affected by anything which interferes with the conduction of sound waves to the cochlea, such as a perforated tympanic membrane (ear drum), disease of the middle ear or disease of the cochlea itself or its connection in the central nervous system.

The sounds reaching the ear are characterized by *loudness (intensity)*, which depends upon the amplitude of the waves; by *pitch*, which depends upon their frequency; and by *quality*, which results from the combination and interaction of the waves. The human ear responds to vibrations ranging from 20 to 20,000 Hz. The waves of speech and many other common sounds are not of single frequency but are complex waves made up of several frequencies of vibration. The number of sound frequencies in addition to the fundamental tone, i.e. the degree of purity of the sound wave is related to the quality or timbre of the sound. The human ear can, in fact, distinguish some 400,000 different sounds.

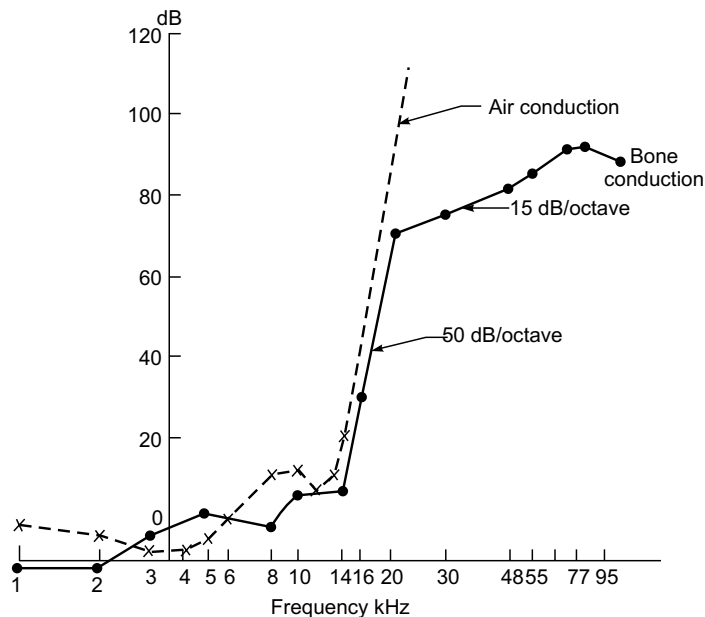
17.1.1 Air and Bone Conduction

Air conduction, by definition, is the transmission of sound through the external and middle ear to the internal ear. Bone conduction, on the other hand, refers to transmission of sound to the internal ear mediated by mechanical vibration of the cranial bones and soft tissues. The most important diagnostic differential from the standpoint of the functional hearing tests is the relationship between air and bone conduction acuity. Clinical observation has shown that hard-of-hearing patients with middle ear disease usually have normal hearing by bone conduction, whereas patients with inner ear involvement have decreased or diminished bone-conduction.

It has been concluded from clinical observations that an approximate 60 dB loss is the maximal air conduction impairment to be anticipated with middle ear defect. Therefore, if the air conduction loss in a patient with apparently typical middle ear pathology exceeds 60 dB, it is likely that inner ear impairment is superimposed on the middle ear lesion.

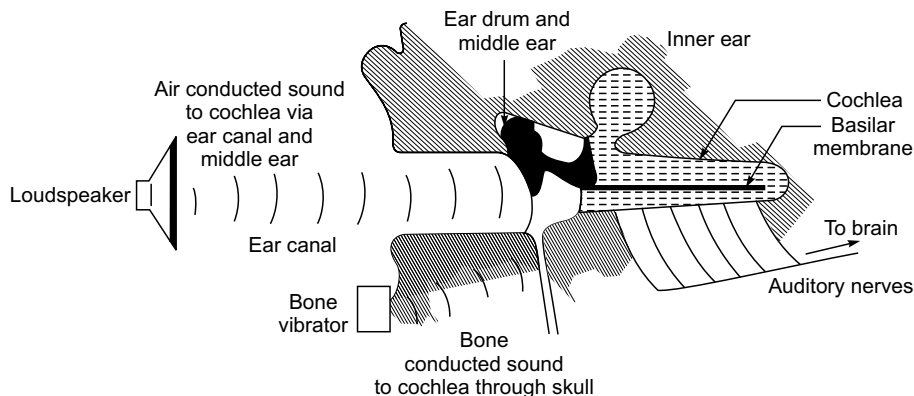
Fig. 17.2 shows that up to a frequency of about 14–16 kHz, the thresholds for air conduction and bone conduction have the same shape. At this frequency, they both show an abrupt fall in sensitivity with a slope of 50 dB/octave. The start of this slope defines the 'end point' of the ear.

For air conducted signals, the fall in sensitivity continues, so that for instance at 25 kHz, 5 W of acoustic power (equivalent to about 500 W of electrical power) is needed to produce a hearing response. On the other hand, for bone conducted signals, there is a change in slope again, at about 2 kHz above the end point. From then on up to 200 kHz, the threshold sensitivity falls at a rate of 10 to 15 decibels per octave. So, in the ultrasonic region, a bone conducted signal of less than one electrical watt is audible.



► Fig. 17.2 Air and bone conduction response curves of a normal ear

There is a rapid drop in impedance of the middle ear at high frequencies and very little of the acoustical energy fed to the ear by air conduction is transmitted to the cochlea. But bone-conducted sound by-passes the middle ear, as is shown in Fig. 17.3. This to some extent explains the different threshold shapes at high frequencies.



► Fig. 17.3 Mechanism of air and bone conduction

17.1.2 Threshold of Hearing

The threshold pressure level of a sound is the lowest level at which an observer can discriminate between the desired sound and the noise background always present in the auditory system. In other words, it is minimal intensity on the stimulus scale, which is just barely adequate to elicit a response. The hearing threshold is not an invariable fixed intensity above which sound is always heard and below which sound is never heard. In fact, the sensitivity of the auditory mechanism is found to vary with interactions between certain physiological, psychological and physical factors. Therefore, the threshold may be regarded as an intensity range within which sound stimuli at or near the statistically determined threshold may or may not be perceived.

The base of each audiological examination is the determination of the hearing threshold. A complete audiogram, one which shows both air and bone conduction threshold bilaterally, is not only a graphic representation of the dB loss at different frequency levels for both air and bone conduction but also indicates the type and location of the hearing impairment. Thus, on the basis of the audiograms, the conduction and perceptive hearing deficiencies can be easily separated from each other.

After establishing the sensitivity of the ears to air bone sounds, it is necessary to know whether the hearing loss is due to a disorder of the hearing organ (cochlea) and connecting nerves with the brain (perceptive loss) or whether the loss arises because of reduced transmission of sound vibrations through the middle ear mechanism itself (conductive loss).

17.2 MEASUREMENT OF SOUND

Sound intensity may be defined as the amount of energy flow per unit time through a unit area perpendicular to the direction of energy flow. It is expressed as watts per square centimetre. However, the common receivers of sound are microphones, which do not measure sound intensity directly. They are sensitive to sound pressure and therefore, it is more pertinent to measure sound in terms of sound pressure which is given in dynes per square centimetre or in microbars (one microbar equals 1 dyne per cm²). Sound pressure, for a given sinusoidal event, is related linearly to both amplitude and frequency. Sound intensity is proportional to the square of sound pressure.

In most acoustic considerations what is of importance is not the absolute magnitude of the intensity but its relative magnitude as compared to some assumed reference standard. The convenient unit for making such comparisons and to express the sound intensity and sound pressure data for all practical purposes, is the decibel (dB). The dB is 1/10 of a larger unit, the bell, named after Alexander Graham Bell. Decibel expresses the logarithm of the ratio between two sound intensities, powers or sound pressures. Since dB is merely a ratio, it is a dimensionless entity.

If I_1 and I_2 are two intensities in watts per square centimetre, then the number of decibels with which they are related can be expressed as:

$$N = 10 \log I_1/I_2$$

$$= 20 \log P_1/P_2$$

where P_1 and P_2 are sound pressures in dynes per square centimetre. Since intensity is proportional to the square of sound pressure, the constant 20 has been used in this case.

Attenuation is commonly expressed in negative dB numbers whereas amplification is given in positive dB numbers.

Use of decibels as unit for comparison of intensities helps to avoid all mathematical calculations except algebraic addition or subtraction of small numbers. The transmission efficiency of any medium like air, a hearing aid or an amplifier is usually expressed in dB as a gain when the output is greater and as a loss, if less. When two media are connected in series, the net efficiency is the sum of the two. For example, if a patient has a hearing loss of 70 dB and is fitted with a hearing aid with a gain of 60 dB, the net efficiency of the air-to-bone transmission system comes out to be $-70 + 60$ or a 10 dB loss.

Level (Volume): This is sound power measured in dB from a chosen zero level of power. The zero level is usually 10^{-16} W per cm^2 . This is the sound power (intensity) in air flowing past an area of 1 sq cm at right angles to the direction of the sound, at the rate of 10^{-16} W. This is approximately the just audible power of a 1000 Hz pure tone for young healthy adults. The corresponding reference value in terms of sound pressure is 0.0002 dyne per sq cm. The smallness of both the sound pressure and sound intensity values show that the human hearing mechanism is very sensitive.

17.2.1 Transducers

In audiometry, generally employed transducers are (i) earphone, (ii) microphones, (iii) electret microphone, (iv) MEMS-based microphone, (v) bone vibrators, and (vi) loudspeakers.

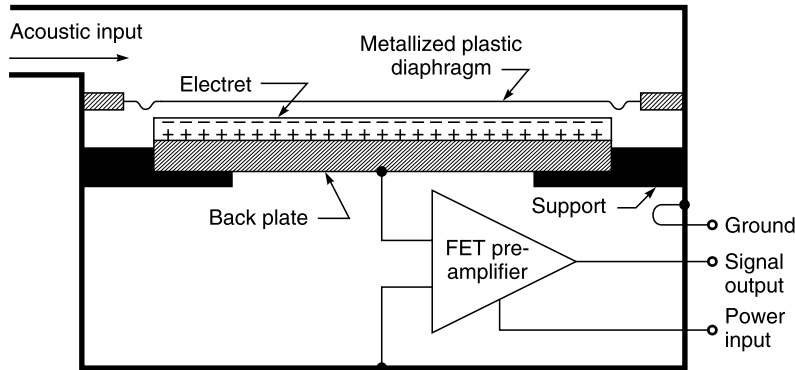
Earphones: They are usually of the moving coil type. They give a reasonably flat frequency response up to 6 kHz after which their sensitivity falls rapidly. They are not specially designed for audiometric applications but for communication purposes. In their miniature form, they are used in hearing aids. When used via insert earphone and ear moulds, they provide greater acoustic power to be transferred to the small volume of the external ear. It may be noted that audiometer earphones are not interchangeable and must remain identified with a specific instrument to preserve its calibration.

In conditions of ambient noise being too high for unshielded earphones, specially designed audio cups are used to exclude background noise which will otherwise result in elevated threshold measurements.

Microphones: These are used to translate wave motion in air into electrical signal. Usual types are: (i) carbon button which changes resistance with air pressure, (ii) electrodynamic where a voltage is induced in a coil by its motion relative to a magnet, (iii) condenser where capacitance of a condenser is varied by the vibration of one of the condenser plates. High quality condenser microphones of diameters 12.5, 6.25 and 3.125 mm are currently used, depending on the frequency to be measured. For special purposes, microphones can be fitted to the ear caps and used in reciprocal arrangement to transmit sound to the ear.

Electret Microphone: Fig. 17.4 shows an electret condenser microphone which consists of a very light diaphragm (moving plate) and back plate (stationary or static plate) and has a permanent charge implanted in an electret material to provide polarizing voltage. The principle of operation is that sound waves impinging on the diaphragm cause the capacitance between it and the back plate to change synchronously, this in turn induces an AC voltage on the back plate. The AC signal is amplified in an amplifier which consists of a single JFET transistor, with the gate connected to the pick-up plate, the source connected to ground, and the signal appearing on the drain. The JFET in this electret microphone is designed for low-noise applications.

MEMS-based Microphone: Analog Devices, Inc. has developed a high performance MEMS microphone (ADMP801) specifically for hearing aid applications. When compared to legacy



► Fig. 17.4 Integrated microphone and FET preamplifier (Electret microphone)

solutions like electret condenser microphones, the ADMP801 is not only smaller in size at only 7.3 cubic millimeters, but also offers greater performance stability over time, temperature, and environmental changes, produces very low equivalent input noise (EIN) at 27 dBA SPL (sound pressure level), and consumes only 17 μ A at 1V supply -- a fraction of the power consumed by traditional microphones. The ADMP801 MEMS microphone is available in a tiny surface-mount package measuring only 3.35 mm $\times$ 2.50 mm $\times$ 0.98 mm, thus offering advantages in terms of noise performance, package size, and phase and gain stability. Fig.17.5 shows the relative size of MEMs based microphone for hearing aids with respect to the size of the ear.

Bone Vibrators: In the early days, bone vibrators were composed of a rod connected to an electromagnetically excited driving system. They were often held by hand against the mastoid process. Such units were large and cumbersome and are not in clinical use today. In the present form, bone vibrators are of the hearing-aid type in which the transduction mechanism changes the alternating current into a vibratory force through a diaphragm. The diaphragm and its basic mechanical parameters like mass, compliance and resistance are important in establishing its response characteristics. Though convenient, it is a very inefficient means of transduction and has a rather limited and peaky frequency response. The plane circular contact area of a bone vibrator is recommended to be $175 \pm 25 \text{ mm}^2$. It is held in position by a headband.

Loudspeakers: They are used to deliver auditory stimuli, when it is not possible to have close coupling of the transducer to the ear. Obviously, the acoustic energy loss into the surroundings will be much greater than when stimulation is applied directly via an earphone. The acoustics of the test room and masking of the non-test ear are two important factors which merit special considerations while using the sound field of a loudspeaker.

17.3 BASIC AUDIOMETER

An audiometer is a specialized equipment, which is used for the identification of hearing loss in individuals, and the quantitative determination of the degree and nature of such a loss. It is



► Fig. 17.5 The relative size of MEMs-based microphone for hearing aids with respect to the size of the ear (Courtesy: Analog Devices)

essentially an oscillator driving a pair of headphones and is calibrated in terms of frequency and acoustic output. Both frequency and output are adjustable over the audio range. The instrument is also provided with a calibrated noise source and bone-conductor vibrator.

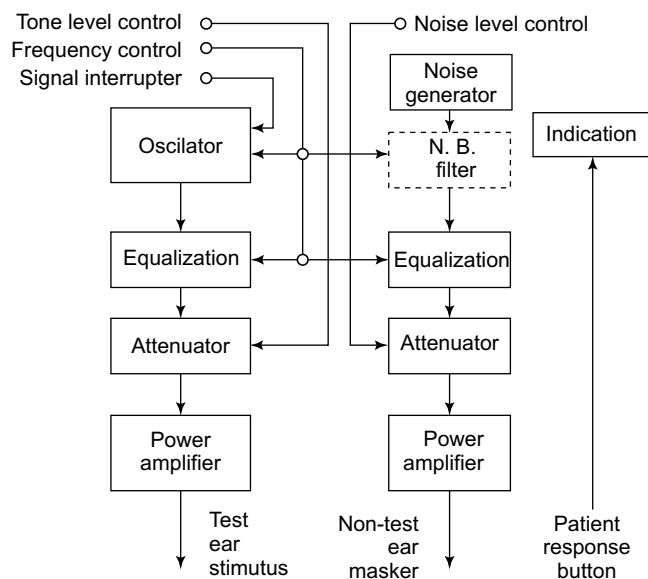
Audiometers may be divided into two main groups on the basis of the type of stimulus they provide to elicit auditory response: pure-tone audiometers and speech audiometers. A pure-tone audiometer is used primarily to obtain air-conduction and bone-conduction thresholds of hearing.

These thresholds are helpful in the diagnosis of hearing loss. Pure-tone screening tests are employed extensively in industrial and school hearing conservation programmes. Speech audiometers are normally used to determine speech reception thresholds for diagnostic purposes and to assess and evaluate the performance of hearing aids.

Screening audiometers are used to separate two groups of people. One that can hear as well as or better than a particular standard and the other that cannot hear so well. Applications of these instruments are found in industry, schools and military service.

In conventional pure-tone audiometry, head phones are worn by the subject and a set of responses is obtained for air-conducted sounds directed to each ear in turn. A bone conductor vibrator can then be attached to the head at the centre forehead position to see whether the hearing threshold improves. If it does, then the disorder is most likely wholly or partly conductive in origin. To avoid stimulation of the ear not under test with the vibrator, it can be temporarily made deaf by introducing a suitable masking noise in the non-test ear via an earphone. A narrow-band noise centred on the pure-tone test frequency or a wide-band white noise is used for this purpose. The problem of how to recognize the need for masking and then applying the correct intensity poses a considerable difficulty.

Fig. 17.6 shows a block diagram of a basic audiometer. It consists of two channels, namely tone generator and noise generator, and each channel having an attenuator, equalization circuit, and power amplifier. The tone generator or oscillator has a controllable frequency range from



► Fig. 17.6 Block diagram of a basic audiometer (Adapted from Patel and Pandey, 2002)

250 Hz to 8 kHz. The equalization circuit is required to provide frequency dependent attenuation in order to calibrate the output sound levels in dB HL and also to provide different amount of attenuation for different output devices used (headphone, loudspeaker, and vibrator). The attenuator, known as the hearing or tone level control, should be capable of controlling the output sound level over a desired range in steps of 5 dB. The output sound level should be within 3 dB of the indicated value.

For the masking purpose, the noise generator should provide wide band noise, which has energy spectrum equally distributed over the test frequency range i.e. up to 8 kHz. There is a facility for narrow band noise, whose output should be distributed around the test frequency. The output power available from the power amplifier determines the maximum sound pressure level available from the headphones and the bone vibrator. The amplifier must have low distortion and a good S/N ratio to meet the standard requirements (Nalamwar and Pandey, 2004).

17.3.1 General Requirements of Audiometers

Modern audiometers are solid-state instruments covering a frequency range from approximately 100 to 10,000 Hz. Some instruments produce this range in discrete octave or semi-octave steps or intervals, while others provide for continuously variable frequency over their designed range. The frequency must remain sensibly constant at a value within 1–3% of the indicated value. Where automatic recording facilities include a continuous sweep frequency, the rate of change is normally kept as one octave per minute. If an automatic recording audiometer provides fixed frequencies, then a minimum period of 30 s must be allowed at each frequency.

The test frequencies should have sufficient purity of tone or approximation to the ideal sine wave form to ensure response only to the desired fundamental frequency. The maximum harmonic distortion in pure-tone air conduction audiometry is specified as 2% for the second and third harmonic and much less at higher order harmonics. The total harmonic distortion should not be more than 3%.

The intensity range of most audiometers starts from approximately 15 dB above normal to 95 dB below normal over a frequency range from approximately 500 to 4000 Hz. The intensity range is somewhat less for frequencies below 500 Hz and above 4000 Hz. This is partly because of certain instrumental limitations imposed by the earphone or vibrator and partly due to the desire to avoid the threshold of feeling from stimulation at the lower frequency levels. The threshold of feeling is the sensation of pain or tickle in the ear, which results from sound pressures and limits the maximal sound intensity that can be tolerated by the ear. The intensity level at which the threshold of feeling is stimulated varies with frequency. For example, the threshold of feeling is stimulated at an intensity level approximately 120 dB above the normal threshold of audibility from about 500 to 4000 Hz, but at 64 Hz the threshold of feeling is stimulated by sound pressures approximately 65 dB above the normal threshold value.

The attenuation dials on the audiometers provide variable intensity or volume controls. They are calibrated in decibels usually in discrete steps, which differ by 5 dB in intensity from step to step. Auditory acuity for each frequency is thus measured in dB above or below the normal hearing-zero dB reference level for that frequency. This level is the minimal intensity at which each given frequency can be perceived by the normal ear in a noise free environment and is experimentally determined by averaging the results of measurement on a large number of normal individuals between 18 and 25 years of age.

In the recent years, numerous audiometric products incorporating microprocessors have been introduced in the market. Such equipment offers greater convenience in calibration, test signal presentation and versatility. Automated data collection and storage are also useful features included in such equipment. It may however be noted that the audiometric measurement principles as described in the following text are not altered with the use of microprocessors and digital technology.

It is extremely important in audiometry to ensure that only the testing signal reaches the ear. Therefore, all testing must be done in a noise-free environment. Since environmental noise is difficult to control, the noise-free conditions are achieved by performing the audiometric testing in a sound isolating enclosure. Such enclosures help to attenuate all frequencies within the sensory range below the threshold of hearing of normal ears. By using double wall construction and appropriate sound absorbing material, it is common to achieve 25 dB attenuation at 125 Hz and 60 dB attenuation at frequencies between 1000 and 8000 Hz.

17.3.2 Masking in Audiometry

In the presence of monaural and asymmetrical binaural hearing losses, there is serious difficulty in obtaining accurate measures of hearing for the poorer ear. The answer to the problem is to eliminate responses from the better ear by masking in order to shift the threshold to a higher level, permitting greater intensities to be presented to the poorer ear without any danger of cross-over. If the difference in air conduction acuity between the two ears is 50 dB or more, then it is advisable to place a masking noise over the better hearing ear while determining the threshold in the other. Masking efficiency depends upon the nature of masking sound as well as its intensity. A pure tone can be used to mask other pure tones. However, over a range of test frequencies, masking efficiency of a pure tone is low as compared to a noise composed of many frequencies, as usually provided in commercial audiometers.

Saw-tooth noise and white noise have been most commonly used for masking in clinical audiometry, but narrow band noise, i.e. a restricted frequency bandwidth of white noise is also often used. Saw-tooth noise is a noise in which the basic repetition rate (fundamental frequency) is usually that of the mains voltage and contains only those frequencies that are multiples of the fundamental. The intensity of these multiples decreases as their frequencies increase. Noises referred to as 'complex' or square waves are similar in that they are composed of a fundamental frequency and components that are multiples of it.

White noise is a noise containing all frequencies in the audible spectrum at approximately equal intensities. However, the spectrum is limited at the ear by the frequency response of the earphone, which may essentially be flat to 6000 Hz and may drop rapidly beyond. An excellent complex masking noise can be obtained by using the thermal or random electronic emission from a semiconductor diode, since it generates all frequencies simultaneously and with equal amplitude over a frequency range wider than the response of the ear.

Narrow-band noise has been used by a number of investigators in audiometric studies. It is produced by selectively filtering white noise. It has been found that narrow band noise is the most efficient masking noise in pure-tone audiometry. The masking audiograms for normal hearing subjects and the clinical results for hearing-impaired subjects show that for equal intensity levels, narrow band noise produces greater threshold shifts than do either of the other two types and thereby provides greater protection from false responses due to cross-over of the test tone.

17.4 PURE-TONE AUDIOMETER

A wave in air, which involves only one frequency of vibration, is known as pure-tone. Pure-tone audiometry is used in routine tests and, therefore, it is the most widely used technique for determining hearing loss. Pure-tone audiometers usually generate test tones in octave steps from 125 to 8000 Hz, the signal intensity ranging from -10 dB to $+100$ dB. .

Pure-tone audiometry has several advantages, which makes it specifically suitable for making threshold sensitivity measurements. A pure-tone is the simplest type of auditory stimulus. It can be specified accurately in terms of frequency and intensity. These parameters can be controlled with a high degree of precision. Speech audiometry normally allows measurements to be made within the frequency range of 300–3000 Hz. Some patients may have impaired high frequency response due to high intensity level occupational noise at 4000 or 6000 Hz. Pure-tone measurements at these frequencies prove to be a more sensitive indicator of the effect of such noise on the ear than speech tests. Changes in threshold sensitivity associated with various middle ear surgical procedures can be monitored more accurately with pure-tone than speech tests.

A pure-tone audiometer basically consists of an oscillator for having a precise control on the frequency of oscillations. The oscillator is coupled to an output current amplifier stage to produce the required power levels. The attenuators used in these instruments are of the ladder type, of nominal $10\ \Omega$ impedance. The signals are presented acoustically to the ear by an earphone or small loudspeaker. The available sound pressure levels in a typical audiometer are given in Table 17.1.

• Table 17.1 *Test Tones and Signal Intensity in Audiometers*

Frequency	Pure-tone (head-phones)	Pure-tone (bone conduction)	Balance channel	Narrow band masking (head-phones)	Narrow band masking (bone conduction)
125	70	-	-		
250	90	45	90	80	50
500	110	60	110	90	60
1000	110	60	110	90	60
1500	110	60	110	90	60
2000	110	60	110	90	60
3000	110	60	110	90	50
4000	110	60	110	90	50
6000	90	-	-	80	-
8000	90	-	-	80	-
Speech	110	-	110	-	-

17.5 SPEECH AUDIOMETER

Besides tonal audiometry, it is sometimes necessary to carry out tests with spoken voices. These tests are particularly important before prescribing hearing-aids and in determining the deterioration of speech understanding of patients. Specially designed speech audiometers are used for this purpose. They incorporate a good quality CD player, which can play recorded speech. Masking noise is supplied by the noise generator. The two channels supply the two head-phones or the two loudspeakers which are of 25 W each.

The CD player has a capacity for recording a limitless variety of test material and a consistency of speech input, which cannot be obtained for live-voice audiometry in relation to test-retest repeatability. Another advantage of the recorded material is that the test words and sentences can be selected to cater for the widely differing needs of age, intelligence, dialect and language.

In speech audiometers, live-voice facilities are incorporated primarily for communication purposes as the inherent unreliability of live-voice speech tests may lead to serious errors. The microphone amplifier used for this purpose is a simple two stage amplifier. The frequency response characteristics of a live-voice channel should be such that with the microphone in a free sound field having a constant sound pressure level, the sound pressure level developed by the earphone of the audiometer in the artificial ear at frequencies in the range 250 to 4000 Hz does not differ from that at 1000 Hz by more than 110 dB. Also, it shall not rise at any frequency outside this band by more than 15 dB, relative to the level at 1000 Hz.

17.6 AUDIOMETER SYSTEM BEKESY

George Van Békésy, a Hungarian scientist, designed an automatic audiometric testing method for plotting the hearing threshold based on the patient's signal. This is another form of pure tone audiometry wherein a self-recording audiometer is used in which the changes in intensity as well as frequency are done automatically by means of a motor. A principal feature of the method, differentiating it from conventional pure-tone audiometric techniques, is the interdependence of the patient's response and stimulus intensity: responses govern intensity and are affected by changes they introduce in it. An audiogram traced by the Békésy method represents the absolute threshold values at all frequencies in the range tested. In addition, it shows the difference, in decibels, between levels at which the patient just hears a signal of increasing intensity and those at which he just ceases to hear the signal when its intensity is decreasing. This latter characteristic often varies significantly with the type of hearing impairment, and can aid in establishing the site of lesion within the auditory system. On the basis of the audiograms, one can easily separate the conduction and perceptive hearing deficiencies from each other.

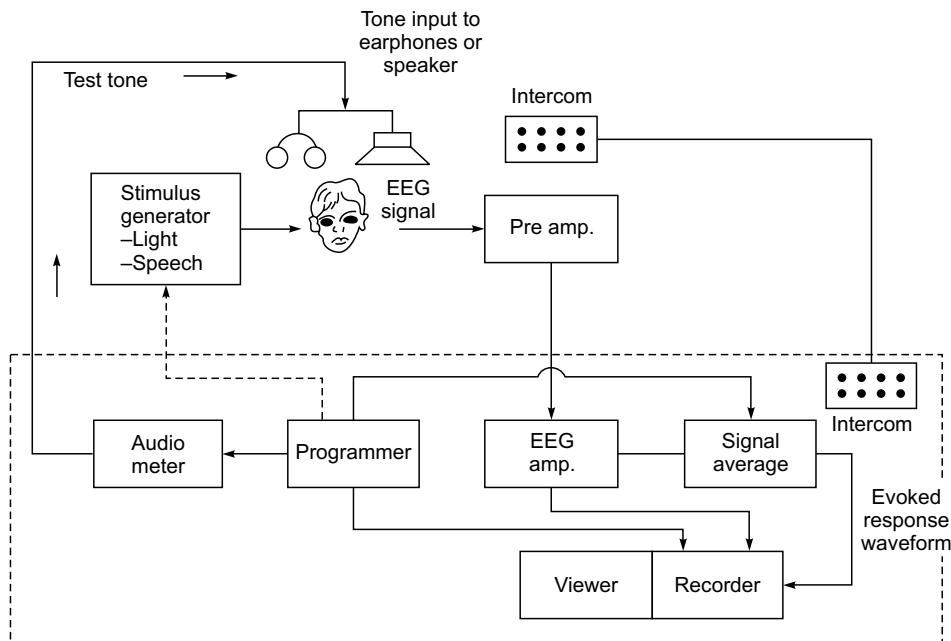
Békésy Audiometers are relatively simple for the patient to operate. The instrument generates a pure-tone signal, which is presented to him through an air-conduction earphone. The subject is told to press a switch when the tone is heard and to release the switch when it is not heard. This switch controls the motor-driven attenuator of the audiometer: when it is pressed, signal intensity decreases and when it is released, signal intensity increases. A pen connected to the attenuator traces a continuous record of the patient's intensity adjustments on an audiogram chart, producing a graphic representation of the subject's threshold. A graphical representation of the subjects hearing threshold across the entire frequency range is thus obtained by the successive crossing and re-crossing of the hearing threshold in the form of a jagged line. Two tracings are recorded for each ear, one by presenting a continuous tone and other by presenting a pulsed tone. The test signal may be presented in a variety of ways, each suited to the investigation of a particular problem.

17.7 EVOKED RESPONSE AUDIOMETRY SYSTEM

Evoked response audiometry has been the subject of research for several years. This work has established evoked response electroencephalography resulting from an auditory stimulus above the hearing threshold. Instruments based on this principle have been found particularly suitable for determining auditory threshold in the absence of voluntary response in subjects such as infants, uncooperative adults, or animals.

The system basically comprises a conventional wide range pure-tone audiometer, which operates under the control of an automatic programmer and provides a series of auditory stimuli to the subject via either a loudspeaker or standard earphones. The EEG signal is picked up by standard electrodes placed in contact with the subjects scalp. One electrode is usually placed on the vertex, one at the post auricular area, and a third (ground) on the earlobe or forehead. The instrument stores and evaluates that part of the EEG signal, which follows each individual stimulus presentation. At the end of the programmed series of stimuli, it writes out on a paper chart a waveform that is the average response to stimuli. The presence of characteristic amplitudes and latencies in this waveform give an indication that the test intensity exceeded the subject's threshold at the test frequency. Similar trials at other intensity levels and other frequencies establish the threshold contour.

Fig. 17.7 shows a block diagram of the evoked response audiometry system. It consists of the following five major subsystems.



► Fig. 17.7 Block diagram of the evoked response audiometer

The Tone Generator: It is a wide range pure-tone audiometer whose frequency output can be selected at 250, 500, 750, 1000, 1500, 2000, 3000, 4000, 6000 and 8000 Hz. The output power levels are adjustable from -5 to $+110$ dB in 5 dB steps. In addition to the pure tones, internally generated broad-band noise may be used as the stimulus. Provision is also made for external input from other types of stimulus. A special feature of the generator is the selectable rise/decay times for 1–100 ms. Outputs are provided for the left ear, right ear, or both. A variable intensity masking noise source is included in the generator. A power amplifier is incorporated to drive a speaker or tactile transducer. Total harmonic distortion should not exceed 2%.

EEG Amplifier: It is a conventional high gain, high impedance, low noise amplifier. The first stage of the EEG amplifier is preferably kept in a separate “preamp head” located near the

subject. Its design and location minimize power line frequency pick-up. An ohm meter provided in the preamp head enables the measurement of the electrode contact impedance and thus indicates when satisfactory contact resistance of the electrodes is obtained. The EEG preamplifier gain is fixed at 200. The overall sensitivity is adjustable in steps of 10 to 1000 mV/div on the chart recorder. The amplifier also provides selectable roll-offs at the high and low ends of the spectrum of interest and a 60 Hz sharp notch filter.

The Programmer: A logic device that controls the system operation in correct time sequence. It helps to have a selectable rate of stimulus presentation, stores the number of pulses that the operator chooses to constitute a run, starts the recorder at the beginning of the run, turns the audiometer tone generator on and off to provide the auditory stimuli at the proper time. It also speeds up the chart drive for the detailed signal samples, stops the recorder after providing for paper clearance, erases the signal averaging computer and clears and resets itself for the next run.

Signal Averaging Computer: This separates evoked responses from the normal EEG activity by ignoring those components, which are not synchronized with stimuli. Because the waveform of the evoked potential will be essentially the same every time in response to the tone presentation, and the other electrical activity will vary randomly, the evoked response “grows” in the computer memory and the noise component tends to average to zero with repeated presentations. It may be provided with either 50 or 100 averaging points depending upon the degree of resolution required. The computer includes a provision for selection of integrating time constant and sweep duration or analysis time. A delay circuit is incorporated to select a delay between the onset of the stimulus and the start of analysis by the signal averager. The amplitude of the evoked response may be normalized to that of the on-going EEG monitoring signal using a gain control. The computer provides outputs for display either on a conventional LCD Monitor or on an X-Y plotter.

Chart Recorder: It is a two-channel recorder. One of the two channels is used to display the averaged response after it has been processed by the computer and the other displays unprocessed EEG. There are two event markers, one of which is activated by each gating pulse from the programmer to show the beginning and duration of each stimulus and the other is available for registering any mark at the desired instant. The chart can be driven at four different speeds (1, 5, 25, 125 mm/s), which are automatically switched by the programmer.

Evoked response audiometer systems also contain a provision for ‘External’ mode of operation where any other type of stimulus generator can be connected into the system and controlled from the programmer. This could include narrow band noise, speech, a high frequency auditory signal for animal research or even a photic or tactile stimulator.

Modern evoked response audiometers are built around microcomputers. In these instruments, the stimulators, preamplifiers and amplifiers are all digitally controlled via the central processing unit, which automatically avoids undesirable parameters. They have no push button or dials as the parameters are varied by means of the keyboard. The parameters can be controlled to a very wide range, which would not be possible with conventional knobs and switches. Stimuli from 12 to 16 kHz are included to facilitate investigation into high frequency hearing loss and ototoxic drug effects. A built-in chart recorder helps to make recordings under the control of the keyboard.

■■■■► 17.8 CALIBRATION OF AUDIOMETERS

The purpose of audiometric testing is to compare. The comparison may be between an individual's present audiogram and one taken previously, or between his audiogram and the audiograms

of others. Whatever the reason for recording an audiogram, clearly the test conditions and the reference levels for the measurements must be identical for valid comparisons to be made between them. Furthermore, for valid comparisons with audiograms from other instruments, the test conditions and reference levels must be universal. Various standard institutions have published audiometric standard threshold values for several widely used combinations of earphones and artificial ears for carrying out audiometer calibrations.

Accurate calibration of audiometers is essential to ensure that the instrument produces a pure tone at the specified level and frequency, and that the signal is present only in the transducer to which it is directed. For pure tone and speech audiometers, the parameters, which are commonly checked, are frequency and intensity.

The frequency output from the audiometer is checked using either a counter-timer or an oscilloscope. A counter-timer is preferable because it gives a quick, direct reading. It is connected across one earphone, and the audiometer is set to give maximum output. The specifications for audiometers allow a tolerance of $\pm 3\%$ for frequency output from a fixed frequency pure-tone audiometer.

Sound pressure levels are best checked by an 'artificial ear' or coupler together with a sound level meter. The artificial ear consists of a condenser microphone and a 6 ml coupler. The coupler originally designed by the National Bureau of Standards, USA encloses a volume approximately the same as the volume under a earphone for a human ear. The earphone is placed squarely on the coupler with a 500 g weight. It should be ensured that it forms a good seal. The output on the sound level meter is read directly in dB and compared with the expected output per frequency. It may, however, be noted that volume displacement is only one component of acoustic impedance and, therefore, it is recognized to be a rather crude acoustic approximation of the average human ear. This is why the reference levels vary with the earphone type. The disadvantage of different reference levels is overcome with a more elaborate design of the wide-band artificial ear. A typical example of an artificial ear is that of type 4153 Artificial Ear from Bruel and Kjaer. It is designed in accordance with IEC R 318. It has a three cavity coupler and provides acoustic impedance, which closely resembles that of the human ear. Different types of couplers are available for use with artificial ears for measurements on headphones, insert type hearing aids and other earphones.

While the artificial ear is used as a standardized substitute for the human ear when making measurements using an air-conduction earphone, an artificial mastoid is used as a standardized substitute for the human mastoid. The artificial mastoid consists of a mechanical simulation of the human one, incorporating a built-in force transducer to monitor the output of the device to be calibrated. The artificial mastoid must meet standard specifications such as the American National Standard ANSI S3 13-1972 or the IEC R 373.

Audiometric tests should be conducted in rooms which are reasonably quiet. A subject under test should not be disturbed either by sounds created within the room or by those intruding from the outside. Such rooms are known as sound-treated rooms. In order to keep out extraneous noises, the outside walls of such rooms consist of heavy hard-surfaced shell. The inside is lined with absorptive material to keep reverberation low and to prevent practically any reflection.

17.9 HEARING AIDS

Hearing loss has many forms. The most common is related to the body aging process and to long-term cumulative exposure of the ear to sound energy. As one grows older, it becomes more difficult to hear. The ear becomes less sensitive to sound, less precise as a sound analyser and less effective as a speech processor. Loss of hearing differs greatly in different individuals.

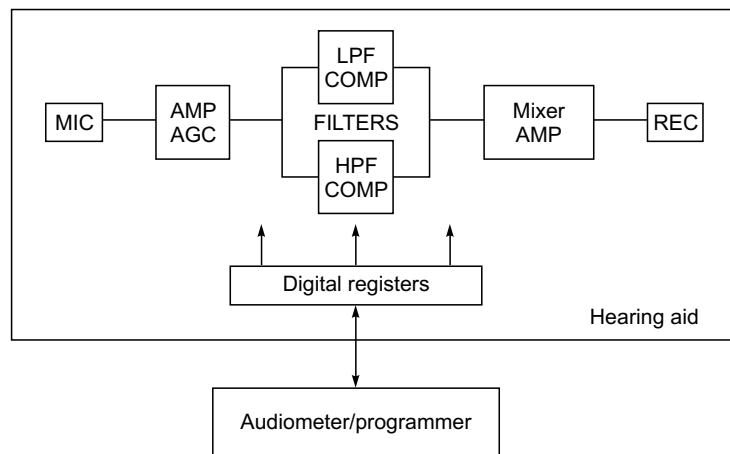
Changes in the ear occur gradually over time. However, by the time the changes are manifested, it is estimated that approximately 30 to 50 percent or more of the sensory cells in the inner ear have suffered irreparable structural damage or are missing (Engebretson, 1994). Under these conditions, the only choice available for hearing-impaired individuals is to wear a hearing aid.

Hearing impairment is caused by either loss in sensitivity (loss in perceived loudness), or loss in the ability to discriminate different speech sounds or both. Loss of loudness may be due to either increased mechanical impedance between the outer ear and the inner ear or by the reduced sensitivity of the sensory organ of hearing. Loss of the discrimination ability is basically associated with damage to the sensory organ, although, other neural structures at higher levels may also be involved.

The modern hearing aid became possible with the invention of the transistor, which has enabled to develop small, power-efficient amplifier circuits that could be packed in a form that fits behind or in the ear. Even though the primary function of a hearing aid is to compensate for the loss of sensitivity of the impaired ear, in practice, it is not this simple. The ear behaves differently for soft sounds near the hearing threshold than it does for loud sounds. Therefore, a frequency response that restores normal hearing thresholds for soft sounds will not, in general be appropriate for louder sounds. Furthermore, even when speech sounds are made audible for the hearing-impaired listener, it does not follow that he/she will be able to understand speech. Hearing-impaired listeners experience more difficulty in understanding speech in background noise than normal- hearing listeners.

17.9.1 Conventional Hearing Aid

Modern hearing aids have evolved from single-transistor amplifiers to modern multi-channel designs containing hundreds and even thousands of transistors. A typical design is shown in Fig. 17.8. The basic functional parts include a microphone and associated preamplifier, an automatic gain control circuit (AGC), a set of active filters, a mixer and power amplifier, an output transducer or receiver. The total circuitry works on a battery. The use of multiple channels in this design provides different compression characteristics for different frequency ranges. Typically, the crossover frequencies of the channels and the compression characteristics can be adjusted



► Fig. 17.8 Conventional analog type hearing aid

with potentiometers. Most of the latest hearing aids are electronically programmable. The programmable parameters are downloaded from a computer-based system and stored in digital registers. The register outputs are used to switch resistor networks that control various analog circuitry. The active filters are adjusted to generally provide for low-frequency attenuation of up to 30–40 dB relative to the high-frequency response. This is because most hearing aid wearers require high frequency gain.

The transducer in a hearing aid, which is an electret microphone, can be realized in an integrated form with a field-effect transistor preamplifier. The preamplifier is housed in the metallic, microphone case to shield its input from extraneous noise. On the other hand, the receiver is an electromagnetic device, which drives a miniature diaphragm to produce acoustic output. The acoustic output is routed to the ear-mould through a flexible tubing whose frequency response can be altered to boost the high-frequency response. This is done by tapering its inside diameter from the ear mould back to the receiver port end.

All the electronics circuitry is packaged in a housing, which can be designed for fitting to the ear in any one of the following ways:

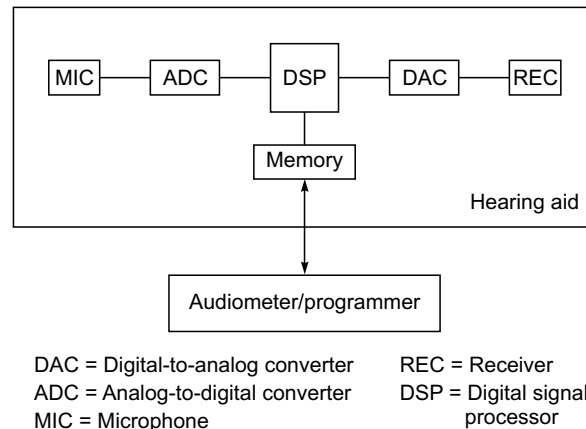
- Placing all the components in a pocket-sized enclosure or box which is connected to the output transducer worn in the ear. The box can be carried in the shirt pocket or carried with a belt around the waist. With the availability of miniature-sized aids, this approach is no longer employed.
- The components are packaged in a curved module, which is designed to fit comfortably behind the ear.
- The most popular design is in which the total package can be put inside the outer ear.

Alternatively, analog programmable hearing aids have a microchip that is programmed for different listening environments. Program settings depend on the individual's hearing-loss profile, understanding of speech, and range of tolerance for louder sounds. Even with the improvement that analog programmable offer, 25.3% of analog hearing aid users reported that they have a hard time listening in presence of high background noise (www.intechopen.com).

Much depends on the performance of the filters for further reduction of the size and improvement in the working of the hearing aids. The dynamic range of an operational amplifier, which is the basic building block of an electronic filter, decreases as the three halves power of feature size. Since dynamic range of analog hearing aids is already marginally acceptable, it appears that further reduction in size to achieve increased processing complexity is not practical. The potential for greater dynamic range, with less power consumption and greater complexity in hearing aid design is feasible only with digital processing technologies.

17.9.2 Digital Hearing Aid

A typical digital hearing aid is illustrated in Fig.17.9. The major parts are the microphone, an analog-to-digital converter (ADC), the digital signal processor (DSP), the digital-to-analog converter (DAC), the receiver and a two port memory. Essentially, sound waves picked up by the microphone and transformed into electrical signals are converted into digital form by an A-D converter. A typical microphone will have an internal noise of 20 dB SPL (sound pressure level) when referred to the input and maximum undistorted output corresponding to a signal of about 90 dB SPL. Allowing some margin for peak performance, the total dynamic range required of the ADC is 80 dB. This requirement can be achieved with a 14 bit A–D converter.



► Fig. 17.9 Block diagram of a digital hearing aid (after Engebretson, 1994)

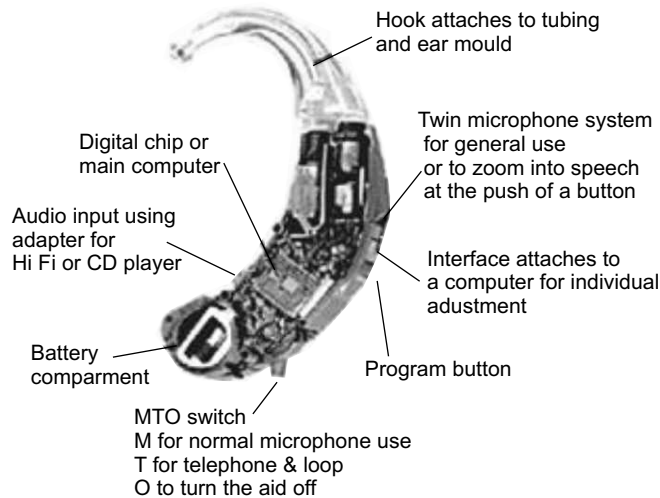
The DSP is a fixed (wired-program) digital processing device containing an array of adders, multipliers and registers which provide the fundamental operations necessary for implementing various digital algorithms. In a general-purpose DSP, considerable power is consumed in executing the programme instructions. Since power consumption is a major consideration in the design of hearing aids, the wired-program approach is followed. The DSP is associated with a two-port memory, which is used to store processing parameters that can be down loaded from the external programmer to the hearing aid while it is adjusted for the intended user.

The hearing aids based on DSP chip use digitized sound-processing algorithms to convert sound waves into digital signals. Key benefits of these include improvement in programmability, greater precision in fitting, management of loudness discomfort, control of acoustic feedback, and noise reduction. A processor chip in the aid analyzes the signals to determine whether the sound is noise or speech. It then makes modifications to provide a clear, amplified, distortion-free signal (Clopton and Spelman, 2000). A programmable DSP also means that the hearing aid algorithms and features could be customized or changed without changing the hardware. Hearing-aid practitioners could economically experiment with available algorithms on a near real-time basis. It would even be possible to have user-selectable programs for switching to highly processed sound in difficult listening situations or back to traditional, less distorted sound in quiet environments.

The dynamic range requirements of the DAC are more severe. Some hearing impaired listeners have almost normal sensitivity at low frequencies but significantly elevated thresholds at high frequencies. Since the conversion noise generated by the DAC has a uniform spectrum and is a function of the overall output signal level, high-level high-frequency sounds can create low-frequency noise and distortion that falls above the threshold at low frequencies.

Overall, the power consumption in current analog and digital hearing aids is approximately equal. Total current consumption is about 0.7 mA to 1.0 mA in the analog devices, whereas digital devices consume 0.5 mA to 0.7 mA. A 1.35-V zinc-air battery that provides around 30 to 65 mAh with a 50- μ A self-discharge current powers this system. The end-of life voltage is about 0.9 V.

The digital hearing aids are implemented with CMOS technology, with a feature size of 1 μ m or less and with an estimated power consumption of 20 μ W. An estimated 10,000 CMOS inverters are required to implement 400,000 multiply-add operations for filtering, compression functions and other processing requirements. Fig. 17.10 shows a typical modern hearing aid.



► **Fig. 17.10** A typical modern digital hearing aid (Courtesy: www.hearingtherapyaustralia.com)

Some of the digital hearing aids in the market use ASICs with programmable coefficients. These ASICs provide a few sets of algorithms and several frequency bands that were not possible in typical analog devices. For example, the digital hearing aids have a combination of the following features: 2 to 14 frequency bands with adjustable crossover frequencies, one microphone, dual microphones for directional listening, background noise reduction, automatic gain control (AGC), speech enhancement, feedback reduction and loud sound protection. Overall, the amount of processing that can be done is impressive, especially when compared with the traditional processing in an analog aid.

The digital hearing aids provide capabilities of superior signal processing, ease of fitting and stable long-term performance. However, potential areas of improvement include shaping the frequency response to invert the patient's hearing loss, enhancing the signal-to-noise ratio with adaptive filtering, reducing acoustic feedback and compressing/expanding signals with minimum distortion.

17.9.3 Cochlear Implants

Sensorineural deafness affects a large number of people throughout the world (Spelman, 1999). The treatment of choice for the sensorineural deaf is the cochlear prosthesis or cochlear implant. A cochlear implant is a prosthetic inner-ear replacement that provides direct electrical stimulation to the inner ear's auditory nerve, allowing for perception of the sensation of sound. These devices are used for patients with severe-to-profound hearing sensorineural loss who cannot be helped with hearing aids.

Sensorineural deafness can be caused either by cochlear damage or by damage within the auditory nerve or to the neurons of the central auditory system. The hair cells are the sensory cells that transduce mechanical motion into signals that can be recognized by auditory neurons. The auditory neurons carry information from the hair cells to the cochlear nucleus in the brainstem and, via the cochlear nucleus, to higher nuclei in the brain (Fig. 17.11).

The normal cochlea and the associated neurons of the central auditory system provide information about both the frequency content and intensity of the auditory signal. Information